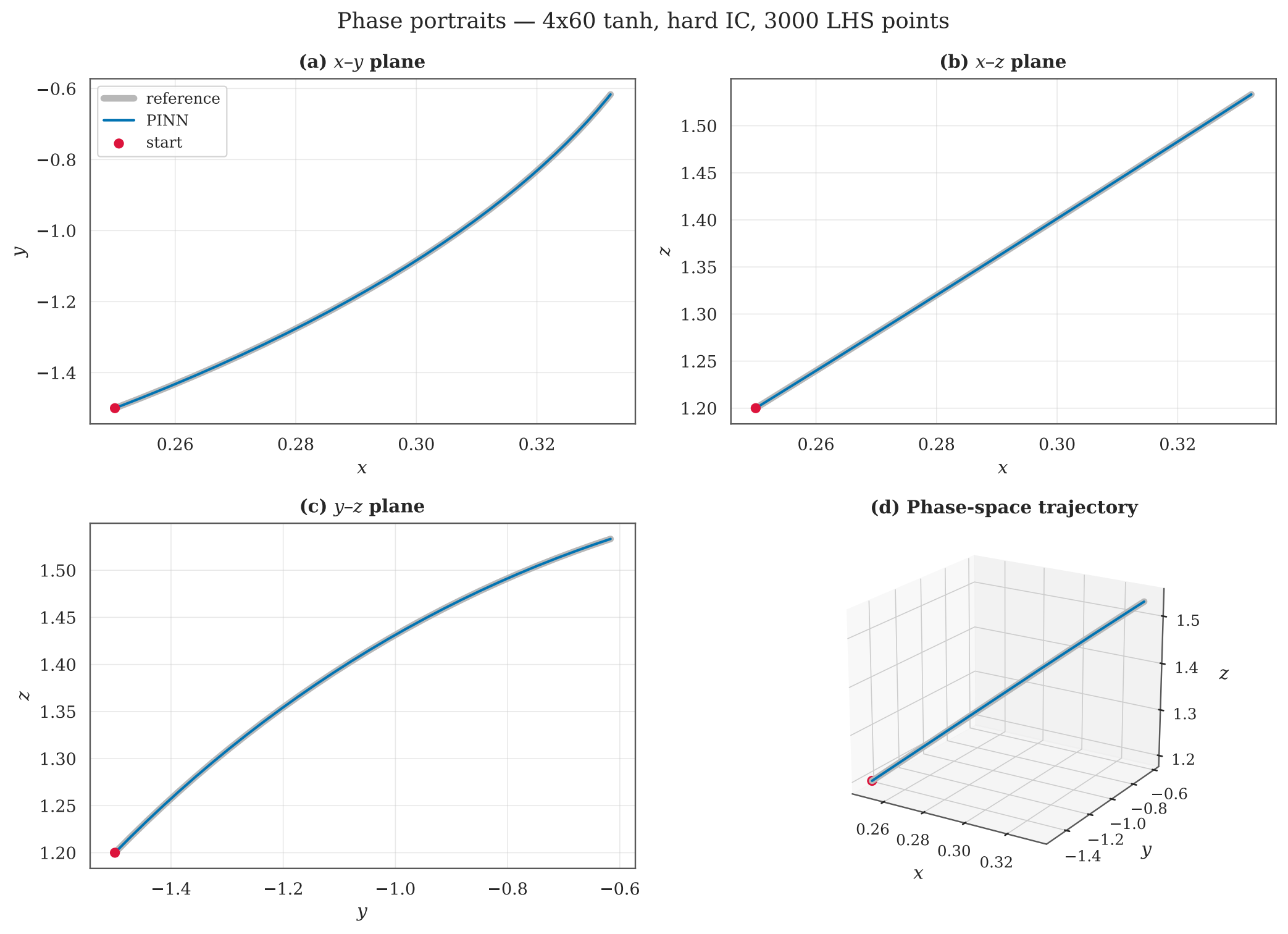
Phase portraits — 4x60 tanh, hard IC, 3000 LHS points
(a) x–y plane
reference
PINN
start
−0.6
−0.8
−1.0
−1.2
−1.4
y
0.26
0.28
0.30
0.32
x
(b) x–z plane
1.50
1.45
1.40
1.35
1.30
1.25
1.20
z
0.26
0.28
0.30
0.32
x
(c) y–z plane
1.50
1.45
1.40
1.35
1.30
1.25
1.20
z
−1.4
−1.2
−1.0
−0.8
−0.6
y
(d) Phase-space trajectory
1.5
1.4
1.3
1.2
z
−0.6
−0.8
−1.0
−1.2
−1.4
y
0.26
0.28
0.30
0.32
x